

Digital Egypt Pioneers Initiative

Bank Loan Risk Analysis

Data Analysis with Power BI Final Project

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1. Project Proposal & Strategic Overview

The core objective of this business intelligence initiative is to analyze loan risk and customer behavior related to loan acquisition. This comprehensive evaluation explores the underlying reasons driving customers to take loans, alongside the multifaceted factors that actively influence their financial decisions. By developing a robust data solution in Power BI, the project extracts meaningful, data-driven insights that empower the bank to systematically identify high-risk or unreliable customers.

Ultimately, these analytical assets enable the institution to execute superior, risk-adjusted decisions regarding loan approvals and overall credit risk management. The deployment of this framework serves to structurally improve financial decision-making and minimize potential credit losses across the bank's lending portfolio.

2. Stakeholder Analysis & Requirement Gathering

- Primary Beneficiaries — Lenders (Banks & Financial Institutions):

As the central stakeholders, lenders utilize the final analytical models to evaluate loan applications objectively. The platform provides the necessary tools to rigorously assess customer risk profile parameters, lower aggregate loan default rates, and establish optimized lending governance.

- Indirect Beneficiaries — Investors:

Investors capitalize on the model's outputs to audit and evaluate the macro-level risk exposure of the bank's active loan portfolio. It provides visibility into loan performance trends and portfolio stability, serving as a vital baseline before capital allocation or investment decisions are finalized.

3. Team Composition & Resource Allocation

To ensure a streamlined execution of the data pipeline, team member responsibilities were strictly segregated into distinct functional domains:

Team Members	Assigned Engineering & Functional Responsibilities
Mohamed Hany & Mrawan Said	Data Cleaning and Preprocessing phase execution.
Malak Elsayed & Shahd Fagelnour	Data Modeling, Schema Design, and Relationship Mapping.
Sandra Bassem & Login Tamer	Data Visualization, UI/UX Dashboard Layout, and Interactive Reporting.

4. Data Lifecycle: Collection & Preprocessing

The engineering workflow adheres to a systematic project plan divided into chronological progression steps:

4.1 Data Collection

The primary data asset for this project was extracted from an established banking and credit repository hosted on Kaggle: <https://www.kaggle.com/datasets/zaurbegiev/my-dataset>.

4.2 Data Cleaning & Preprocessing Pipeline

During this technical stage, the raw dataset underwent rigorous formatting and transformation to secure absolute accuracy, operational consistency, and analytical reliability. The pipeline executed the following automated and manual tasks:

- Promoted Headers: Standardizing the first row of data to serve as system-wide column headers.
- Format Standardization: Ensuring uniform column naming conventions and structural formatting across all tables.
- Data Type Validation: Auditing and correcting field assignments to prevent type mismatch errors during calculations.
- Text Extraction: Parsing, isolating, and truncating descriptive fields to retrieve structured dimensions.
- Value Replacement: Correcting erroneous inputs, handling misspellings, and standardizing categorical strings.
- Irrelevant Data Filtering: Purging columns and records that did not contribute to target risk factors.
- Missing Value Resolution: Evaluating null elements to implement structured structural treatments.

Completion of this phase successfully guaranteed a clean, reliable, and validated data foundation, ready for modeling and downstream analysis.

4.3 Preprocessing Risks & Mitigation Strategy

- Identified Risks: Critical data omissions (such as missing values across key vectors like income, age, and loan amount), duplicate records, and inconsistent field data types.
- Applied Mitigations: Complete elimination of duplicate entries, hard alignment of text/numeric types, and resolving null fields by either programmatic removal or isolating them into dedicated tracking categories within calculated columns.

5. Data Modeling & Architecture

To facilitate high-performance, low-latency analysis within Power BI, the dataset was restructured into an optimized analytical model.

5.1 Star Schema Optimization

The system relies on an industry-standard Star Schema structure. This separation avoids performance overhead and scales effectively:

- Fact Table (fCredit): Holds the core transaction-level records, numeric loan balances, and risk outcomes.
- Dimension Table (dCustomer): Retains distinct customer attributes and descriptive demographics. Generated cleanly by creating a duplicate of the original staging table and removing unnecessary transactional columns to maximize model performance.
- Relationship Parameters: A unidirectional, **one-to-many relationship (Single direction)** was established from dCustomer to fCredit. This mapping reflects the business rule where a single unique customer can actively hold multiple loans over time.

5.2 Data Enrichment: Calculated Columns

A suite of calculated columns was engineered to enrich customer profiling. To maintain dashboard clarity and performance, these columns were hidden from end-user view once structural boundaries were set:

<i>Calculated Column Name</i>	<i>Functional Logic & Operational Purpose</i>
<i>Annual Income Availability</i>	Flags whether a borrower's annual income is explicitly documented or missing.
<i>Annual Income Category</i>	Segments individuals into discrete Low, Average, or High-income tiers.
<i>Credit Score Availability</i>	Audits and marks the presence or absence of a recorded credit bureau score.
<i>Credit Score Category</i>	Groups scores into Fair, Good, Very Good, or Needs Clarification categories.
<i>Adjusted Current Loan Amount</i>	Identifies and normalizes anomalies (e.g., system dummy values like 99,999,999) using a realistic median value.
<i>Debt-to-Income Ratio</i>	Formulated as: $(\text{Monthly Debt} * 12) / \text{Annual Income}$ to establish annualized comparisons.
<i>Debt-to-Income Category</i>	Classifies individuals into Low, Medium, and High-risk tiers based on financial burden.
<i>Loan Amount Category</i>	Partitions total financing volumes into Small, Average, and Large groupings.
<i>Months Since Last Delinquent Category</i>	Categorizes repayment delays into No Delinquency, Recent Delinquency, or Old Delinquency.
<i>Years-in-Job Category</i>	Tiers work history into New Employee, Low Experience, Medium Experience, or High Experience.

5.3 Specialized Measures Table (DAX Metrics)

To support consistent, dynamic aggregations, a dedicated, isolated measures table was implemented to house all DAX calculations:

1. Average Annual Income: Evaluates the average yearly revenue across borrowers to benchmark baseline earning levels.
2. Average Credit Balance: Tracks the mean outstanding credit balance to evaluate current unpaid debt obligations.
3. Average Credit Score: Calculates average credit scores to summarize borrower creditworthiness and portfolio risk.
4. Average Debt-to-Income Ratio: Measures the average relationship between debt and income to gauge repayment capacity.
5. Average Loan Amount: Identifies typical financing scales and capital requests across the loan portfolio.
6. Average Monthly Debt: Computes standard recurring monthly liabilities to isolate direct consumer overhead.
7. Average Year in Current Job: Benchmarks employment longevity to assess professional and income stability.
8. Charged off Loans: Quantifies the total count of default instances where loans were deemed uncollectible bad debt.
9. Charged off Ratio: Formulated as $\text{Charged off Loans} / \text{Total Loans}$ to capture the structural proportion of toxic loans.
10. Default Rate: Computes the percentage of defaulted financing tracks out of aggregate loans to evaluate portfolio risk.

11. Fully Paid Loans: Counts total lending agreements that successfully achieved absolute repayment.
12. Fully Paid Ratio: Represents the percentage of loans that concluded successfully (Fully Paid / Total Loans).
13. Num of Borrowers: Calculates DISTINCTCOUNT(Customer ID) to supply a clean baseline of unique customers.
14. Total Loan Amount: Sums total capital issued (SUM(Loan Amount)) to track macro lending volume.

6. Core Performance Indicators & Portfolio Statistics

The following validated Key Performance Indicators (KPIs) summarize the baseline state of the banking dataset:

1. Average credit score: 716.28
2. Total loan Amount: 32.19
3. Average loan amount: 312.50k
4. Charged off ratio: 22.64%
5. Average debt-to-income ratio: 17.25%
6. Average of bankruptcies: 0.12
7. Average of maximum open credit: 757.53k
8. Average of annual income: 1.38M
9. Average of number of open accounts: 11.12
10. Average of years in current job: 5.88
11. Average of loan amount: 312.5k
12. Num of borrowers: 82k
13. Average monthly debt: 18.5k
14. Average credit balance: 294.65k
15. Number of open accounts: 81.8k
16. Sum of bankruptcies: 9k
17. Sum of Tax liens: 2k

7. Business Intelligence Dashboard Architecture

The interactive report contains 6 interconnected dashboard pages supported by context-aware cross-filtering and global navigation menus.

7.1 Page 1: Summary Dashboard

- KPI Performance Cards: Average Cleaned Score, Total Loan Amount, Average Loan Amount, and Charged-off Ratio.
- Donut Chart (Fully Paid vs. Charged-off Loans): Evaluates clean repayment distributions against critical default rates.
- Treemap (Total Loan Amount by Purpose): Displays capital concentration across user-specified loan categories.
- Bar Chart (Default Rate by Purpose): Isolates and surfaces high-risk loan intentions to guide policy adjustments.

7.2 Page 2: Credit Score Analysis

- KPI Performance Cards: Average Debt-to-Income Ratio, Average Bankruptcies, and Average Maximum Open Credit.
- Gauge Chart (Average Credit Score): Tracks real-time macro portfolio credit health against target indices.
- Scatter Chart (Annual Income vs. Credit Balance): Cross-referenced by Credit Score Category to locate client clusters.
- Bar Chart (Loan Amount by Credit Score Category): Visualizes capital exposure across specific risk classes.
- Funnel Chart (Default Rate by Delinquency Category): Tracks default escalation based on the age of the last delinquency.
- Line Chart (Charged-off Loans by Credit Score Category): Maps the volume of failed agreements against risk bands.

7.3 Page 3: Borrower Profile

- KPI Performance Cards: Average Annual Income, Average Number of Open Accounts, and Average Years in Current Job.
- Gauge Chart (Default Rate): Identifies institutional default exposure against risk tolerance margins.
- Bar Chart (Avg Loan Amount by Years-in-Job & Status): Measures the impact of employment longevity on performance.
- Line Chart (Total Loan Amount by Monthly Debt): Illustrates structural debt correlations.
- Line Chart (Default Rate vs. Num of Borrowers): Analyzes default risk alongside volume by Years-in-Job Category.
- Bar Chart (Average Annual Income by Loan Status): Compares income capacities between healthy and failed accounts.
- Donut Chart (Default Rate by Term): Breaks down the variation of default frequencies between short and long-term loans.

7.4 Page 4: Assets & Housing Overview

- KPI Performance Cards: Average Credit Balance, Average Loan Amount, and Total Count of Borrowers.
- Asset Matrix Analysis: Pivots Home Ownership (Rent, Own, Mortgage) against volume, averages, and balances.
- Bar Chart (Default Rate by Home Ownership): Identifies housing segments carrying elevated risk profiles.
- Treemap (Total Loan Amount & Avg Income): Highlights leading asset segments driving loan volumes.
- Bar Chart (Average Monthly Debt by Home Ownership): Explains differences in financial overhead across segments.
- Line Chart (Customer Count by Credit Score): Flags borrower concentrations across credit score segments.
- Donut Chart (Borrowers by Loan Status): Displays a distribution breakdown of active vs. completed loan tracks.

7.5 Page 5: Financial Burden & Debt Profile

- KPI Performance Cards: Average Monthly Debt, Average Credit Balance, Open Accounts, and Debt-to-Income Ratio.
- Gauge Chart (Average Debt-to-Income Ratio): Monitors consumer financial over-extension in a simple visual.
- Area Chart (Avg Monthly Debt by Delinquency History): Tracks changes in debt based on historical delinquency age.
- Scatter Chart (Avg Credit Score vs. Monthly Debt): Cross-filters by loan status and open account volumes.
- Bar Chart (Average Loan Amount by Monthly Debt): Identifies consumer segments facing higher debt burdens.
- Line Chart (Avg Monthly Debt by Open Accounts & Status): Evaluates compounding liability pressures.

7.6 Page 6: Loan Product & History

- KPI Performance Cards: Average Loan Amount, Sum of Bankruptcies, and Sum of Tax Liens.
- Treemap (Average Loan Amount by Purpose): Compares financing volume distributions like business vs. medical bills.
- Line Chart (Default Rate by Loan Amount Category): Shows how default rates shift across small, average, and large loans.
- Donut Chart (Average Loan Amount by Term): Compares financing size differences between short and long-term structures.
- Public Records Table: Cross-references Average Credit Score against active Tax Liens to isolate risks.
- Risk Matrix (Bankruptcies vs. Tax Liens): Maps compounding public failures to identify high-risk customers.
- Contextual User Tooltips: Implemented across visuals to reveal detailed sub-metrics without crowding the workspace:
 - Tooltip Line Chart: Illustrates links between active tax liens and broader credit problems.
 - Tooltip Bar Chart (Reasons): Isolates the top 5 driving reasons behind customer loan requests.
 - Tooltip Bar Chart (Debt): Compares monthly debt pressures anchored across home ownership groups.

8. Strategic Insights & Credit Policy Recommendations

Based on the empirical patterns extracted from the asset portfolio, the following corporate actions are recommended:

1. Credit Approval Optimization: Given the notable portfolio default rate, the bank must upgrade its underwriting standards and implement more rigorous approval criteria.
2. Debt-to-Income Caps: Customers exhibiting elevated debt-to-income ratios demonstrate clear default tendencies; strict maximum caps must be enforced to limit high-burden exposure.
3. Risk-Adjusted Score Mandates: Applicants falling into lower credit score categories carry significantly higher risk and must be subjected to stringent terms or higher collateral baselines.
4. Income-Centric Repayment Customization: Borrowers in high-income categories should be offered flexible, custom repayment terms, capitalizes on their robust capacity to pay back.
5. Delinquency Auditing: Customers showing historical late-payment or delinquency markers must trigger manual reviews before any conditional approval is granted.
6. Large Exposure Threshold Controls: Capital risk escalates with larger loan amounts; major capital disbursements must carry tighter structural conditions, restrictive covenants, or lower maximum ceilings.
7. Holistic Multi-Factor Risk Matrix: Credit decisions must transition away from single-metric models and rely instead on consolidated multi-factor matrices to capture multi-layered borrower risk.